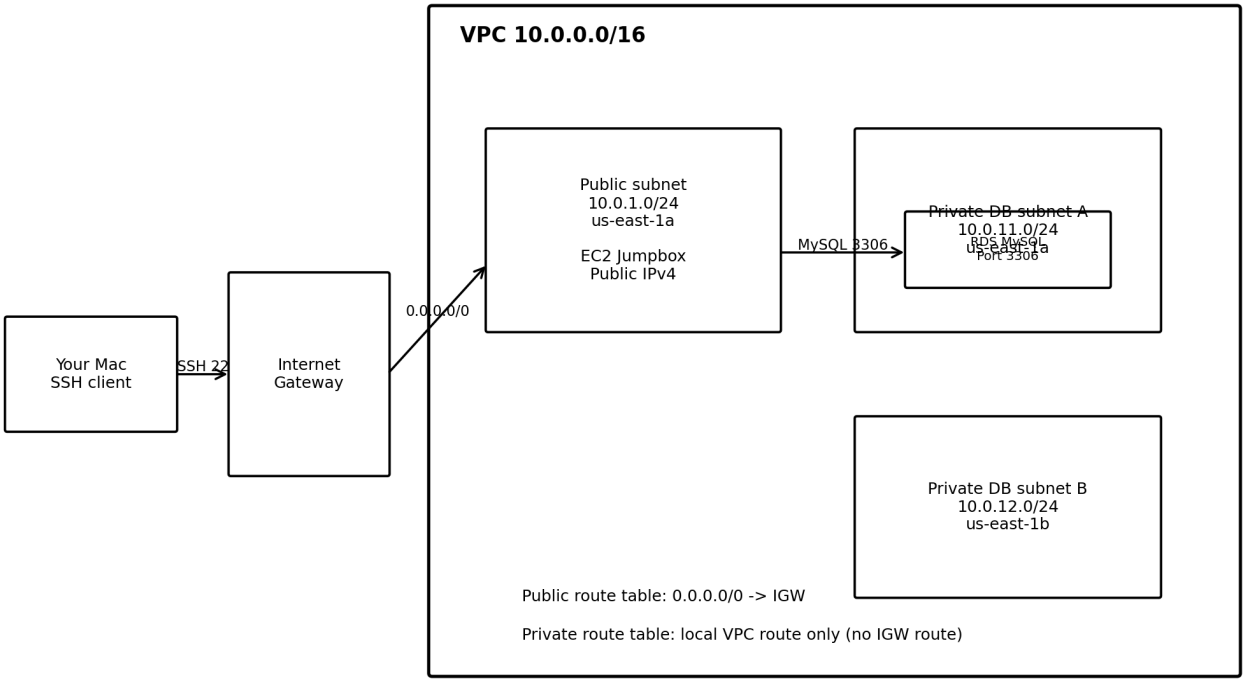
# AWS VPC + EC2 Jumpbox + Private RDS MySQL

## Complete End-to-End Hands-On Lab

Region: us-east-1 (N. Virginia) | Console-first lab with Linux and SQL validation




Goal: build the network from scratch, place a jumpbox in a public subnet, keep RDS MySQL private, connect EC2 to RDS, and prove the database works with SQL.

**Cost warning:** This lab creates billable AWS resources. RDS and EC2 can incur charges. Delete the resources in the cleanup section when finished.

# 1. What You Will Build

• A custom VPC: **10.0.0.0/16**.

• One public subnet in us-east-1a for the EC2 jumpbox.

• Two private database subnets, in us-east-1a and us-east-1b, for the RDS DB subnet group.

• An Internet Gateway and a public route table for the jumpbox subnet.

• A private route table with no Internet Gateway default route for the database subnets.

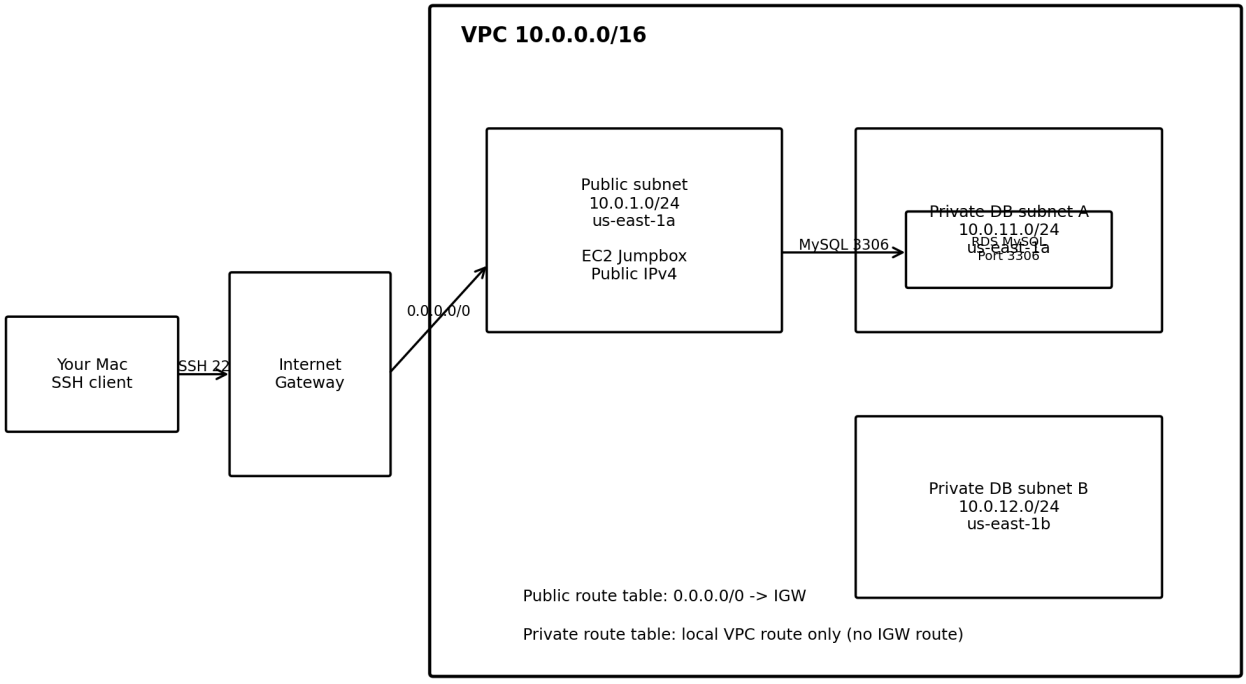
• An EC2 Amazon Linux 2023 jumpbox with a public IPv4 address.

• An RDS for MySQL DB instance with **Public access = No**.

• Separate security groups: SSH to EC2 only from your IP; MySQL to RDS only from the EC2 security group.

• A MySQL client on EC2, a test database/table, INSERT/SELECT/UPDATE/aggregate queries, and connectivity troubleshooting.

## Architecture




# 2. Addressing and Naming Plan

| Resource | Name | CIDR / Port | AZ / Notes |
|---|---|---|---|
| VPC | mysql-lab-vpc | 10.0.0.0/16 | DNS hostnames/resolution enabled |
| Public subnet | public-jumpbox-1a | 10.0.1.0/24 | us-east-1a |
| Private DB subnet A | private-db-1a | 10.0.11.0/24 | us-east-1a |
| Private DB subnet B | private-db-1b | 10.0.12.0/24 | us-east-1b |
| Internet Gateway | mysql-lab-igw | 0.0.0.0/0 route | Attached to VPC |

| Resource | Name | CIDR / Port | AZ / Notes |
|---|---|---|---|
| Public route table | public-rt | 0.0.0.0/0 -> IGW | Associated with public subnet |
| Private route table | private-db-rt | Local route only | Associated with both DB subnets |
| EC2 security group | lab-jumpbox-sg | TCP 22 | Source = your public IP /32 |
| RDS security group | lab-rds-sg | TCP 3306 | Source = lab-jumpbox-sg |
| EC2 instance | mysql-jumpbox | SSH | Amazon Linux 2023 |
| RDS instance | mysql-lab-db | MySQL 3306 | Private; Single-AZ is fine for lab |

# 3. Phase A - Create the VPC

Open the AWS Management Console, select **N. Virginia (us-east-1)**, then open **VPC**. Choose **Your VPCs > Create VPC**. Select the option to create the VPC only so that every networking component in this lab is explicit.

| Setting | Value |
|---|---|
| Name tag | mysql-lab-vpc |
| IPv4 CIDR | 10.0.0.0/16 |
| IPv6 | No IPv6 CIDR block |
| Tenancy | Default |

After creation, select the VPC and verify that **DNS resolution** and **DNS hostnames** are enabled. RDS endpoints are DNS names, so DNS support is important.

# 4. Phase B - Create Three Subnets

Go to **VPC > Subnets > Create subnet**. Choose mysql-lab-vpc and create the following subnets. If your account does not offer exactly us-east-1a/us-east-1b because AZ labels can map differently by account, choose two distinct AZs.

| Subnet | AZ | IPv4 CIDR | Role |
|---|---|---|---|
| public-jumpbox-1a | us-east-1a | 10.0.1.0/24 | Public EC2 |
| private-db-1a | us-east-1a | 10.0.11.0/24 | RDS subnet group |
| private-db-1b | us-east-1b | 10.0.12.0/24 | RDS subnet group |

For **public-jumpbox-1a**, select the subnet, choose **Actions > Edit subnet settings**, and enable **Auto-assign public IPv4 address**. Leave public IPv4 auto-assignment disabled on both private DB subnets.

# 5. Phase C - Internet Gateway and Routing

**Create the Internet Gateway:** VPC > Internet gateways > Create internet gateway. Name it **mysql-lab-igw**. Select it, choose **Actions > Attach to a VPC**, and attach it to mysql-lab-vpc.

**Create public-rt:** VPC > Route tables > Create route table. Choose mysql-lab-vpc. Open its Routes tab, choose Edit routes, and add:

| Destination | Target |
|---|---|
| 10.0.0.0/16 | local (automatic) |
| 0.0.0.0/0 | mysql-lab-igw |

On **Subnet associations**, explicitly associate **public-jumpbox-1a** with public-rt.

**Create private-db-rt:** create another route table in mysql-lab-vpc and name it private-db-rt. Do **not** add a 0.0.0.0/0 route to the Internet Gateway. Associate both private-db-1a and private-db-1b with it.

For this lab, a NAT Gateway is intentionally omitted. The RDS database does not need outbound Internet access to accept a MySQL connection from EC2 inside the VPC.

# 6. Phase D - Create Security Groups

Create both security groups under **EC2 > Network & Security > Security Groups** or the VPC console.

## Jumpbox security group - lab-jumpbox-sg

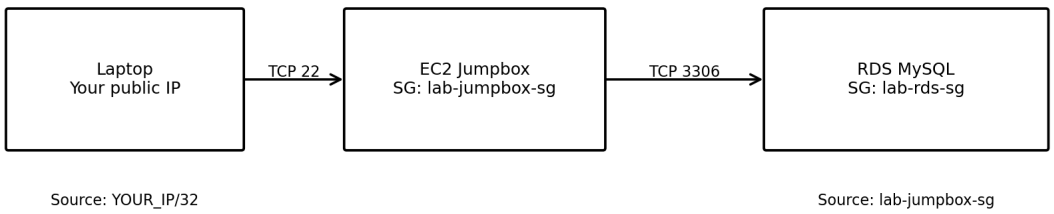
| Direction | Type | Port | Source/Destination |
|---|---|---|---|
| Inbound | SSH | 22 | Your public IPv4 address /32 |
| Outbound | All traffic | All | 0.0.0.0/0 (lab default) |

Use the console's **My IP** selector for SSH. Do not use 0.0.0.0/0 for SSH in this lab.

## Database security group - lab-rds-sg

| Direction | Type | Port | Source/Destination |
|---|---|---|---|
| Inbound | MYSQL/Aurora | 3306 | lab-jumpbox-sg (security group ID) |
| Outbound | All traffic | All | 0.0.0.0/0 (default) |

**Security-group flow**




The RDS inbound rule references the EC2 security group, not the EC2 public IP. Security-group references are a clean way to authorize the workload inside the VPC.

# 7. Phase E - Launch the Public EC2 Jumpbox

Go to **EC2 > Instances > Launch instances**. Use these settings:

| Setting | Value |
|---|---|
| Name | mysql-jumpbox |
| AMI | Amazon Linux 2023 |
| Instance type | t3.micro or another small lab-appropriate type |
| Key pair | Create/select a key pair you possess |
| VPC | mysql-lab-vpc |
| Subnet | public-jumpbox-1a |

| Setting | Value |
|---|---|
| Auto-assign public IP | Enable |
| Security group | Select existing: lab-jumpbox-sg |
| Storage | Default is sufficient for this lab |

Launch the instance and wait for **Running** and the status checks to pass. Record its **Public IPv4 address**.

## Connect from your Mac terminal

```
chmod 400 ~/Downloads/your-key.pem
ssh -i ~/Downloads/your-key.pem ec2-user@EC2_PUBLIC_IP
```

If SSH times out, verify: instance is in public-jumpbox-1a; it has a public IPv4 address; public-rt has 0.0.0.0/0 -> IGW; the subnet is associated with public-rt; and lab-jumpbox-sg allows TCP 22 from your current public IP.

# 8. Phase F - Create the RDS DB Subnet Group

Open **RDS > Subnet groups > Create DB subnet group**.

| Setting | Value |
|---|---|
| Name | mysql-lab-db-subnet-group |
| Description | Private DB subnets for MySQL lab |
| VPC | mysql-lab-vpc |
| Availability Zones | us-east-1a and us-east-1b |
| Subnets | private-db-1a and private-db-1b |

A DB subnet group tells RDS which VPC subnets it may use. Using private subnets in multiple AZs prepares the networking for RDS placement/failover options while keeping the database off the public Internet.

# 9. Phase G - Create RDS MySQL in the Private Subnets

Open **RDS > Databases > Create database**. Choose **Standard create** when available so you can explicitly control networking.

| Setting | Recommended lab value |
|---|---|
| Engine | MySQL |
| Engine version | A currently supported MySQL version offered by the console |
| Template | Free tier if your account/selection is eligible; otherwise Dev/Test |
| DB instance identifier | mysql-lab-db |
| Master username | admin (or another non-reserved admin name) |
| Credentials | Set a strong password; store it securely |
| Instance class | Small lab-appropriate class; use free-tier eligible option if shown |
| Storage | Small/default lab allocation; avoid unnecessary autoscaling for a short lab |
| VPC | mysql-lab-vpc |
| DB subnet group | mysql-lab-db-subnet-group |
| Public access | NO |
| VPC security group | Choose existing: lab-rds-sg |
| Availability Zone | No preference is fine |
| Port | 3306 |
| Initial database name | labdb (if Additional configuration exposes this field) |
| Backup / monitoring | Use modest lab settings; disable extras you do not need if cost is a concern |
| Deletion protection | Off for a disposable lab |

Choose **Create database**. Wait until status is **Available**. Select the DB and copy the **Endpoint** from Connectivity & security. The endpoint resembles `mysql-lab-db.xxxxx.us-east-1.rds.amazonaws.com`. Do not use the ARN as the hostname.

**Checkpoint:** Connectivity & security should show Publicly accessible = No, port 3306, the mysql-lab-db-subnet-group, and lab-rds-sg.

# 10. Phase H - Install a MySQL-Compatible Client on EC2

SSH to mysql-jumpbox. On Amazon Linux 2023, install the available MariaDB/MySQL-compatible command-line client. Start with:

```
sudo dnf update -y
sudo dnf install mariadb105 -y
mysql --version
```

If that exact package name is not available in the current Amazon Linux repository, discover the offered client package and install it:

```
dnf search mariadb
dnf search mysql
```

You only need a client capable of speaking the MySQL protocol; you are not installing a database server on the jumpbox.

# 11. Phase I - Test Network Connectivity Before Login

Set the RDS endpoint in a shell variable. Replace the placeholder with the endpoint copied from RDS.

```
export RDS_ENDPOINT="mysql-lab-db.xxxxxxxxxxxx.us-east-1.rds.amazonaws.com"

getent hosts "$RDS_ENDPOINT"
nc -vz "$RDS_ENDPOINT" 3306
```

A successful TCP test proves that DNS, VPC routing, and security groups permit the connection. If nc is unavailable, proceed directly to the mysql command or install an appropriate networking utility.

Now connect:

```
mysql -h "$RDS_ENDPOINT" -P 3306 -u admin -p
```

Enter the RDS master password only when prompted. Do not put the password directly on the command line because it can be exposed in shell history/process listings.

# 12. Phase J - SQL Validation Lab

At the mysql> prompt, run the following. If you created an initial database named labdb, USE it; otherwise CREATE it.

```
CREATE DATABASE IF NOT EXISTS labdb;
USE labdb;

CREATE TABLE customers (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name  VARCHAR(50) NOT NULL,
    last_name   VARCHAR(50) NOT NULL,
    email       VARCHAR(100) UNIQUE,
    city        VARCHAR(50),
    state       CHAR(2),
    signup_date DATE,
    annual_income DECIMAL(12,2)
);

INSERT INTO customers
(first_name, last_name, email, city, state, signup_date, annual_income)
VALUES
('Alice','Carter','alice@example.com','Arlington','VA','2026-01-15',82000.00),
('Brian','Lee','brian@example.com','Fairfax','VA','2026-02-10',96500.00),
('Carla','Gomez','carla@example.com','Bethesda','MD','2026-03-05',110000.00),
('David','Nguyen','david@example.com','Alexandria','VA','2026-04-22',74000.00);

SELECT * FROM customers;

SELECT state, COUNT(*) AS customer_count,
       ROUND(AVG(annual_income),2) AS avg_income
FROM customers
GROUP BY state
ORDER BY customer_count DESC;
```

```
UPDATE customers
SET annual_income = annual_income * 1.05
WHERE state = 'VA';

SELECT customer_id, first_name, last_name, state, annual_income
FROM customers
WHERE annual_income >= 90000
ORDER BY annual_income DESC;

SHOW TABLES;
DESCRIBE customers;
SELECT DATABASE(), CURRENT_USER(), VERSION();
```

Expected result: four rows after the INSERT; VA has three customers and MD one; the UPDATE increases the VA incomes by 5%; the final diagnostic query identifies the active database, authenticated MySQL user, and engine version.

# 13. Optional - Create a Least-Privilege Application User

Using the RDS master account for applications is poor practice. For learning, create a dedicated user with rights only on labdb. Choose a strong unique password.

```
CREATE USER 'labapp'@'%' IDENTIFIED BY 'REPLACE_WITH_A_STRONG_PASSWORD';
GRANT SELECT, INSERT, UPDATE, DELETE ON labdb.* TO 'labapp'@'%';
SHOW GRANTS FOR 'labapp'@'%';
```

Exit and reconnect as labapp:

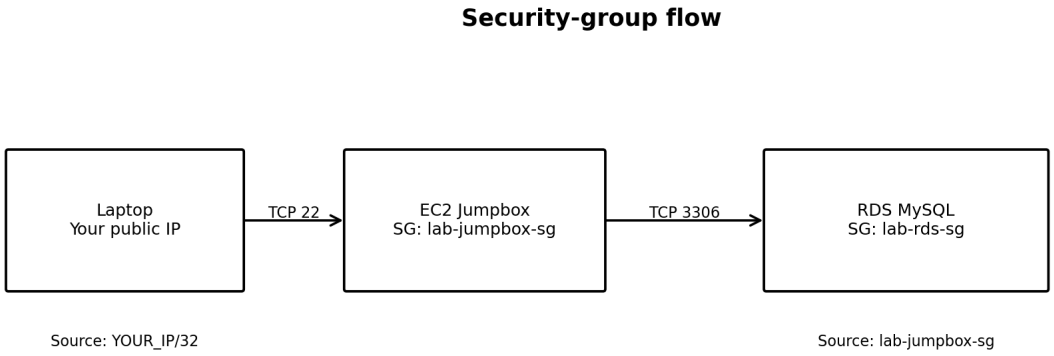
```
exit
mysql -h "$RDS_ENDPOINT" -P 3306 -u labapp -p labdb
SELECT COUNT(*) FROM customers;
```

# 14. Understand the Packet Path

**SSH path:** Laptop -> Internet -> Internet Gateway -> public route table -> EC2 network interface -> lab-jumpbox-sg TCP/22.

**Database path:** EC2 -> VPC local routing -> private subnet/RDS network interface -> lab-rds-sg TCP/3306. This traffic does not need to traverse the Internet Gateway.

**Why the RDS endpoint works:** RDS provides a DNS endpoint. The EC2 instance resolves it to the address RDS currently uses. Applications should connect to the endpoint rather than hard-coding an IP.




# 15. Troubleshooting Matrix

| Symptom | Likely cause / check |
|---|---|
| SSH timeout | Check EC2 public IPv4, SSH inbound YOUR_IP/32, public route table -> IGW, subnet association, instance status. |
| SSH 'Permission denied (publickey)' | Wrong key pair/user/file permissions. Amazon Linux commonly uses ec2-user. Ensure chmod 400 on the private key. |
| RDS hostname does not resolve | Verify VPC DNS support/hostnames; copy the endpoint exactly from RDS. |
| nc/mysql times out on 3306 | RDS SG must allow MYSQL/Aurora 3306 with source = lab-jumpbox-sg; verify both resources are in the same VPC. |
| Access denied for user | Network path is working. Check username/password and MySQL grants. |
| Unknown database 'labdb' | Create it with CREATE DATABASE labdb, or use the initial DB name you configured. |
| Can't choose DB subnet group | Confirm both private subnets belong to mysql-lab-vpc and the DB subnet group was created successfully. |

| Symptom | Likely cause / check |
| --- | --- |
| RDS becomes publicly accessible | Modify the DB and set Public access to No; ensure its subnet/routing/security design remains private. |

# 16. Verification Checklist

• VPC CIDR is 10.0.0.0/16.

• Public subnet is associated with public-rt.

• public-rt has 0.0.0.0/0 -> Internet Gateway.

• Both DB subnets are associated with private-db-rt.

• private-db-rt has no 0.0.0.0/0 -> Internet Gateway route.

• EC2 has a public IPv4 address and lab-jumpbox-sg.

• SSH inbound is limited to your public IP /32.

• RDS shows Publicly accessible = No.

• RDS uses mysql-lab-db-subnet-group and lab-rds-sg.

• lab-rds-sg inbound MySQL/3306 source is lab-jumpbox-sg.

• EC2 resolves the RDS endpoint.

• mysql -h ENDPOINT -u admin -p succeeds.

• SELECT * FROM labdb.customers returns the test rows.

# 17. Cleanup - Avoid Continuing Charges

Delete resources after the lab in roughly this order. AWS may prevent deletion when dependencies still exist.

**1.** RDS > Databases > mysql-lab-db > Actions > Delete. For a disposable lab, you may skip the final snapshot if you do not need the data. Confirm deletion.

**2.** EC2 > Instances > mysql-jumpbox > Instance state > Terminate instance.

**3.** RDS > Subnet groups > delete mysql-lab-db-subnet-group after the DB is gone.

**4.** Delete lab-rds-sg and lab-jumpbox-sg after they are no longer attached.

**5.** Detach and delete mysql-lab-igw.

**6.** Delete the custom route tables and subnets, then delete mysql-lab-vpc.

Also check the EC2 and RDS consoles for any resources you created outside these names before assuming cleanup is complete.

# 18. What This Lab Demonstrates on a Resume / LinkedIn

You can describe the completed project as:

**AWS Networking & Database Lab:** Designed a custom multi-AZ VPC with public/private subnet segmentation, Internet Gateway and route tables; deployed an Amazon Linux EC2 jumpbox in a public subnet and Amazon RDS for MySQL in private DB subnets; implemented least-privilege security-group connectivity for SSH and MySQL; installed a SQL client and validated end-to-end database access with DDL, DML, aggregation, and troubleshooting tests.

A strong next extension is to replace direct SSH exposure with AWS Systems Manager Session Manager, add Secrets Manager for credentials, and deploy a small Python application on EC2 that performs CRUD operations against RDS.

# 19. AWS Documentation Used for Design Validation

This lab follows AWS guidance that commonly places EC2 in a public subnet and RDS in a private subnet, keeps the DB non-public, and uses separate security groups so the database accepts MySQL traffic from the EC2 workload rather than the

Internet. AWS also documents custom VPC/subnet/route-table creation and RDS DB subnet groups for this architecture.

Useful AWS documentation titles to search in docs.aws.amazon.com:

- Tutorial: Connect an Amazon EC2 instance to an Amazon RDS database
- Creating and connecting to a MySQL DB instance
- Tutorial: Create a VPC for use with a DB instance (IPv4 only)
- Option 3: Manually connect an instance to an RDS database by creating security groups
- Creating an Amazon RDS DB instance